

STADIOalot Returns Prediction

Module: Capstone Project

Assignment: SS1 – Part C

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1. Purpose of the Data Request

The purpose of this data request is to collect the information needed to understand what is contributing to product returns at STADIOalot and to build a machine learning model that can predict whether an item is likely to be returned.

STADIOalot already has useful historical data across orders and transactions, products, returns, deliveries, customer shopping behaviour and marketplace sellers. For this project, these different datasets will need to be brought together so that we can look for patterns that may explain why certain items are returned more often than others.

The datasets should be able to connect using common identifiers such as Order_ID, Order_Item_ID, Customer_ID, SKU and Seller_ID. Customer information should be anonymised because information such as names, phone numbers, email addresses and exact home addresses is not needed for this project.

2. Orders and Transactions Data

The orders and transactions data will form the main starting point of the project. It will show what was purchased, when it was purchased, what the customer paid and other information about the order. STADIOalot has six years of orders and transaction history available.

Column name	Data type	Example	Additional information
Order_ID	String	ORD100245	Unique identifier for each order.
Order_Item_ID	String	ITEM100245-01	Unique identifier for each individual item within an order.
Customer_ID	String	CUST58291	Anonymised unique customer identifier.
SKU	String	SKU458210	Unique product identifier that can be linked to the product catalogue.
Seller_ID	String	SELL1025	Identifies the seller responsible for the item.
Order_Date	Datetime	2026-05-14 14:35:20	Date and time the order was placed in YYYY-MM-DD hh:mm:ss format.
Quantity	Integer	2	Number of units purchased. Must be greater than 0.
Unit_Price	Numeric/Decimal	599.99	Actual unit price paid in South African rand.
Listed_Price	Numeric/Decimal	699.99	Normal listed price before a discount, where available.

Discount_Amount	Numeric/Decimal	100.00	Difference between the listed price and actual price paid in ZAR.
Promotion_Applied	Boolean	True	True if a promotion or discount was applied.
Promotion_Type	String/Categorical	20% Off	Type of promotion used, where applicable.
Basket_Value	Numeric/Decimal	1249.97	Total value of the customer's order in ZAR.
Basket_Item_Count	Integer	3	Total number of items contained in the order.
Payment_Method	String/Categorical	Credit Card	Payment method used for the transaction.

This information will help us test whether things such as price, promotions, basket size and the timing of an order are linked to a higher chance of a return.

3. Product Catalogue Data

Product information is important because the STADIOalot briefing pack shows that certain products and certain size and colour combinations are returned more often than others. STADIOalot has six years of product catalogue information available.

Column name	Data type	Example	Additional information
SKU	String	SKU458210	Unique product identifier matching the orders dataset.
Product_Name	String	Men's Running Shoe	Name or description of the product.
Product_Category	String/Categorical	Clothing	Main product category.
Product_Subcategory	String/Categorical	Men's Shoes	More detailed product classification where available.
Brand	String/Categorical	Example Brand	Product brand where available.
Size	String/Categorical	UK 9	Product size where applicable.
Colour	String/Categorical	Black	Product colour or variant where applicable.
Product_Attributes	String	Slim Fit	Other useful product attributes supplied with the listing.
Listed_Price	Numeric/Decimal	699.99	Normal listed selling price in ZAR.

Seller_ID	String	SELL1025	Identifier linking the product to its seller.
Average_Rating	Numeric/Decimal	4.2	Average customer rating where available. Use the platform's normal rating range.
Review_Count	Integer	153	Number of customer reviews available at the time of purchase where possible.
Listing_Age	Integer	365	Number of days the product listing had existed at the time of purchase, if available or able to be calculated.

The original product category values should also be kept because the briefing pack mentions that product categorisation can be inconsistent. This will allow the data quality problem to be identified and handled properly during preparation.

4. Returns Data

Returns data is one of the most important datasets because it provides the outcome that the machine learning model will try to predict. STADIOalot has five years of returns information available.

Column name	Data type	Example	Additional information
Return_ID	String	RET78215	Unique identifier for each return.
Order_ID	String	ORD100245	Links the return to the original order.
Order_Item_ID	String	ITEM100245-01	Identifies the exact item that was returned.
Customer_ID	String	CUST58291	Same anonymised customer identifier used in the orders data.
SKU	String	SKU458210	Product that was returned.
Seller_ID	String	SELL1025	Seller responsible for the returned item.
Return_Date	Datetime	2026-05-20 10:15:00	Date and time the return was recorded.
Return_Reason_Code	String/Categorical	SIZE	Standard return reason code where available.
Return_Reason	String/Categorical	Incorrect size	Description of why the item was returned.

Item_Condition	String/Categorical	Unopened	Condition of the item when it was returned.
Refund_Outcome	String/Categorical	Full refund	Outcome of the refund process.
Refund_Amount	Numeric/Decimal	599.99	Amount refunded to the customer in ZAR.

During data preparation, a new variable called Returned can be created. Returned = 1 would mean the item was returned within the agreed return period, while Returned = 0 would mean it was not returned. This will become the target variable used to train the prediction model.

5. Customer Purchase and Return History

The briefing pack says that customer buying patterns may also be linked to returns. Most of the information below can be calculated from the orders and returns data instead of asking the client for a completely separate customer dataset.

Column name	Data type	Example	Additional information
Customer_ID	String	CUST58291	Anonymised identifier matching the orders dataset.
Customer_Tenure	Integer	730	Number of days since the customer's first recorded order.
Previous_Orders	Integer	27	Number of orders placed before the current order.
Previous_Items_Purchased	Integer	43	Total number of items purchased before the current order.
Previous>Returns	Integer	6	Number of items previously returned.
Historical_Return_Rate	Numeric/Decimal	0.14	Previous returns divided by previous items purchased. Expected range 0 to 1.
Previous_Category_Orders	Integer	8	Previous purchases from the same product category.
Previous_Category>Returns	Integer	2	Previous returns from the same product category.

Average_Order_Value	Numeric/Decimal	745.50	Customer's historical average order value in ZAR.
Days_Since_Last_Order	Integer	21	Number of days between the current order and the customer's previous order.

These variables must only use information that was known before the order being predicted. Future purchases or returns should not be included because that would give the model information it would not have in a real situation.

6. Marketplace Seller Data

Seller information is important because STADIOalot has many marketplace sellers and the briefing pack says that seller quality can vary a lot. Some sellers are linked to more complaints, late shipments and other problems. STADIOalot has five years of seller performance data available.

Column name	Data type	Example	Additional information
Seller_ID	String	SELL1025	Unique seller identifier.
Seller_Rating	Numeric/Decimal	4.3	Historical seller rating at the time of the order where possible.
Seller_Total_Orders	Integer	12540	Number of completed orders handled by the seller before the current order.
Seller_Return_Count	Integer	2257	Historical number of returned items linked to the seller.
Seller_Return_Rate	Numeric/Decimal	0.18	Historical returns divided by items sold. Expected range 0 to 1.
Average_Shipping_Time	Numeric/Decimal	2.4	Historical average shipping time in days.
Complaint_Count	Integer	86	Complaints recorded against the seller before the current order.
Late_Shipment_Rate	Numeric/Decimal	0.07	Historical proportion of late shipments. Expected range 0 to 1.
Seller_Tenure	Integer	1095	Number of days the seller has been active on the platform.

Where seller performance changes over time, the value that was known at the time of the order should be used so that future information does not influence an older prediction.

7. Delivery and Logistics Data

Delivery information can help us understand whether the delivery experience is linked to returns. STADIOalot has four years of logistics information including delivery times, attempts, routes and outcomes.

Column name	Data type	Example	Additional information
Order_ID	String	ORD100245	Links the delivery information to the original order.
Order_Item_ID	String	ITEM100245-01	Include where delivery information is recorded at item level.
Distribution_Centre	String/Categorical	DC_CPT01	Distribution centre that fulfilled the order.
Dispatch_Date	Datetime	2026-05-15 08:20:00	Date and time the order left the distribution centre.
Promised_Delivery_Date	Datetime	2026-05-16 17:00:00	Promised delivery date or time where recorded.
Delivery_Date	Datetime	2026-05-16 13:45:00	Actual delivery date and time.
Delivery_Region	String/Categorical	Western Cape	General delivery area. Exact customer address is not required.
Delivery_Attempts	Integer	1	Number of delivery attempts required.
Delivery_Outcome	String/Categorical	Delivered	Final delivery outcome.
On_Time_Delivery	Boolean	True	Indicates whether delivery happened within the promised delivery period.
Fulfilment_Type	String/Categorical	Own Fleet	For example own fleet, courier partner or another available category.

Some delivery information only becomes available after an order has already been placed. It can still be useful when analysing what causes returns, but it should not be used in a model that is meant to predict return risk before the order is confirmed. This will help avoid giving the model information that would not actually be available at prediction time.

8. Clickstream and Search Data

STADIOalot also records searches, product views, add-to-basket actions, purchases and abandoned sessions. This data could help us understand what a customer did before making a purchase and whether some of those behaviours are linked to returns. Three years of clickstream and search history is available.

Column name	Data type	Example	Additional information
Session_ID	String	SES10582	Unique browsing session identifier.
Customer_ID	String	CUST58291	Anonymised customer identifier where the customer was logged in.
Session_Date	Datetime	2026-05-14 14:20:00	Date and time of the browsing session.
Search_Query	String	black running shoes	Search entered by the customer.
SKU	String	SKU458210	Product interacted with during the session.
Product_View	Boolean	True	Indicates whether the product was viewed.
Added_To_Basket	Boolean	True	Indicates whether the product was added to the basket.
Purchase_Completed	Boolean	True	Indicates whether the interaction resulted in a purchase.
Session_Abandoned	Boolean	False	Indicates whether the session ended without a purchase.

Where possible, the browsing session should be linked to the order so that customer behaviour before the purchase can be connected to the item that was eventually purchased.

9. Data Period Requested

I would preferably request at least three years of overlapping historical data across the different sources. This gives enough information to look for patterns over time while also matching the shortest history available, which is the three years of clickstream and search data. Orders and product catalogue data have six years available, delivery data has four years, and returns and seller data have five years.

10. Data Quality and Preparation Requirements

Before modelling starts, the data needs to be checked carefully. This is especially important because the STADIOalot briefing pack already points out some data quality challenges such as inconsistent product categories, messy address information and seller-supplied fields that may not always be reliable.

- Unique identifiers: Order_ID, Order_Item_ID, Customer_ID, SKU and Seller_ID should stay consistent across the different datasets so that the data can be joined correctly.

- Missing values: Missing information should stay identifiable instead of automatically being replaced with made-up or default values.
- Duplicate records: Duplicate orders, products and return records should be identified.
- Date formats: Dates should preferably use a consistent YYYY-MM-DD hh:mm:ss format.
- Categorical values: Return reasons, product categories and other categories should use consistent values or codes where possible.
- Monetary values: Prices, discounts and refunds should clearly be recorded in South African rand.
- Historical accuracy: Values such as seller ratings and historical return rates should reflect what was known at the time of the order where possible.
- Privacy: Customer identifiers should be anonymised and unnecessary personal information should not be included.

11. Expected Outcome

Once the datasets have been received, they can be joined, cleaned and explored to understand which products, categories, sellers, customer buying patterns and order characteristics are linked to higher return rates.

The prepared data can then be used to train and compare machine learning models that predict the likelihood of an individual order item being returned. The aim is not to stop customers from buying products. Instead, the prediction could help STADIOalot identify higher-risk orders or problem areas earlier and support actions such as better sizing information, clearer product information or further investigation into products and sellers with unusually high return rates.

The final outcome should help STADIOalot use the data it already collects to better understand avoidable returns and support decisions that could reduce unnecessary return-related costs.